



Module Title

Cybersecurity and Networking

Internship Assignment Submission

Wireshark Fundamentals & Packet Analysis

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1.THEORY

1.1 Answer all 20 questions in your own words

1.What is Wireshark?

Wireshark is a network protocol analyzer used to capture and inspect network packets.

2. What is packet capture?

Packet capture means recording network traffic so it can be analyzed later.

3. What is a capture interface?

Packet capture means recording network traffic so it can be analyzed later.

4. Name the three main Wireshark panes.

Packet List

Packet Details

Packet Bytes

5. What is the difference between source and destination IP?

Source IP is where the packet comes from, while destination IP is where the packet is going.

6. What does the Protocol column show?

It shows the network protocol used by the packet, such as TCP,UDP,DNS or ICMP.

7. What is a Wireshark display filter?

A display filter is used to show only specific packets from an existing capture.

8. Which filter shows ICMP traffic?

`icmp`

9. Which filter shows DNS traffic?

`Dns`

10. Which filter shows TCP traffic?

`Tcp`

11. What does `tcp.port == 443` display?

It shows TCP traffic using port 443, which is commonly used for HTTPS.

12. What is DNS?

DNS translates domain names, such as `example.com`, into IP addresses.

13. What is an ICMP Echo Request?

It is a message sent by a device to check whether another device is reachable, such as when using ping.

14. What is an ICMP Echo Reply?

It is the response sent back to the device that sent an ICP Echo Request.

15. What is the TCP three-way handshake?

It is the process TCP uses to establish a connection: SYN- SYN-ACK – ACK.

16. What are SYN, SYN-ACK and ACK?

SYN: Requests to start a connection

SYN-ACK: Accepts the request and responds

ACK: Confirms the connection

17. Why is HTTPS content normally not visible as plain text?

HTTPS uses TLS encryption, so the web content is encrypted and normally cannot be read as plain text in Wireshark.

18. What is a PCAP/PCAPNG file?

It is a file format used to save captured network packets so they can be opened and analyzed later.

19. Why is Wireshark useful in incident response?

It helps security teams investigate suspicious traffic, connections, DNS activity, protocols, and network events.

20. Why must packet capture be performed only on authorized traffic?

Because network traffic can contain private or sensitive information. Capturing someone else traffic without permission can violate privacy and security rules.

2.Lab Practical

2.1 SCREENSHOTS (your own capture only):

Wireshark interface · icmp filter · dns filter · tcp filter · tcp.port == 443 · Packet Details pane · saved day16-lab.pcapng file







